

Array Questions

Question 1:

Given an array of integers, calculate the average of its elements.

Examples:

Input: arr = [2, 4, 6, 8]

Output: 5

Explanation: $(2 + 4 + 6 + 8) / 4 = 20 / 4 = 5$.

Input: arr = [5, 15]

Output: 10

Explanation: $(5 + 15) / 2 = 20 / 2 = 10$.

Question 2:

Find the smallest element in an array.

Examples:

Input: arr = [4, 2, 7, 1, 9]

Output: 1

Explanation: The minimum element is 1.

Input: arr = [5, 5, 5]

Output: 5

Question 3:

Given an array, copy its elements into another array.

Examples:

Input: arr = [5, 10, 15]

Output: newArr = [5, 10, 15]

Explanation: All elements copied to new array.

Input: arr = [1]

Output: newArr = [1]

Question 4:

Check if a given array is a palindrome (same forward and backward).

Examples:

Input: arr = [1, 2, 3, 2, 1]

Output: Yes

Explanation: The array is same from front and back.

Input: arr = [1, 2, 3]

Output: No

Explanation: Forward = [1, 2, 3], Backward = [3, 2, 1]. Not equal.

Question 5:

Find the missing number in a given range from 1 to n.

Examples:

Input: arr = [1, 2, 4, 5], n = 5

Output: 3

Explanation: Range is 1 to 5, missing number is 3.

Input: arr = [2, 3, 1, 5], n = 5

Output: 4

Question 6:

Search an element in an array using linear search.

Examples:

Input: arr = [5, 3, 8, 6], target = 8

Output: Found at index 2

Explanation: arr[2] = 8 matches target.

Input: arr = [1, 2, 3], target = 4

Output: Not Found

Question 7:

Left rotate the array by one position.

Examples:

Input: arr = [1, 2, 3, 4, 5]

Output: [2, 3, 4, 5, 1]

Input: arr = [7, 8]

Output: [8, 7]

Question 8:

Find the difference between the largest and smallest element in an array.

Examples:

Input: arr = [10, 4, 7, 15]

Output: 11

Explanation: Largest = 15, Smallest = 4, Difference = $15 - 4 = 11$.

Input: arr = [2, 2, 2]

Output: 0

Question 9:

Find the second largest element in an array.

Examples:

Input: arr = [10, 20, 4, 45, 99]

Output: 45

Explanation: Largest = 99, Second largest = 45.

Input: arr = [1, 2]

Output: 1

Explanation: Largest = 2, second largest = 1.

Question 10:

Move all zeroes to the end of the array while maintaining the order of non-zero elements.

Examples:

Input: arr = [0, 1, 0, 3, 12]

Output: [1, 3, 12, 0, 0]

Input: arr = [1, 0, 2, 0]

Output: [1, 2, 0, 0]